CPRI PROCEDURE FOR CARBON COMPOSITION OF INSULATING OIL BY FTIR METHOD

(Guidelines from IS: 13155-1991 reaffirmed 2001)

Using a syringe, fill 0.1mm path length cell of the spectrophotometer with oil (ensure transmittance values lie between 20% - 80% T).

Record the infrared spectrum in two regions, (1750-1500) cm^{-1} and (850-600) cm^{-1}.

Draw tangents across the transmission maxima on the either side of the peak being measured. In the case of aromatic carbon, draw tangent between 1650 and 1550 cm^{-1} and in the case of paraffinic carbon, draw tangent between 760 & 680 cm^{-1}.

Use the intensities of the peaks at 1600 cm^{-1} and 720 cm^{-1} to calculate the aromatic carbon C_A and paraffinic carbon C_P respectively, using the formulae:

$$C_A = 1.2 + \frac{9.3 \log I_0/I}{cd}$$

$$C_P = 29.9 + \frac{6.6 \log I_0/I}{cd}$$

Where I_0 = Intensity at the intersection of the tangent & vertical line through the peak minimum.

I = Intensity at the intersection of the vertical line with the peak minimum.

d = Pathlength of cell in mm.

$$= \frac{(n \times 10)}{2(v_2 - v_1)}$$

where n= whole number of fringes occurring between two known wave numbers (v_2 & v_1) (for empty cell)

c= concentration factor (1 for undiluted oils)

Report C_A & C_P to the nearest integer.

Naphthenic carbon, $C_N = 100 - (C_A + C_P)$

Accepted the above procedure:

Name: ______________________________ Designation: ____________________

Organization: ______________________________ Signature: ____________________